CHE 205, Fall 2019, Problem Set 5

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Due date: 11:59pm, Sun, 10/13/2019

All problems are from the textbook by V.L. Law.
Submit your solutions as one file on blackboard. Do all parts of each problem on one tab. Make separate tabs for different problems.

Problem 1 (6 pts). Exercise 4.5.

- In part a, for numerical derivatives, use central differencing and second-order correct approximations for intermediate and end points, respectively. Also, for comparing the numerical derivatives at each point with the exact values, create a column and calculate the percent error.
- In part b, instead of calculating the percent error at $x = 1$, create a column and calculate the percent error for all points. Note that the exact integral is:

$$\int x^2 \cos(x) dx = 2x \cos(x) + (x^2 - 2) \sin(x). \tag{1}$$

Problem 2 (6 pts). Exercise 4.4.

- In part a, use Simpson's rule instead of trapezoidal rule. To experiment with Δx, use $\Delta x = 0.1$ and 0.01 only. In each case, for comparison of results, only calculate the percent error using exact solution (2/3).
- In part b, in each case, for comparison of results, only calculate the percent error using exact solution (2/3).

Problem 3 (6 pts). Exercise 4.1.

- In part b, for curve fitting, use a linear function. Also, for comparing the results with those of part a, only compare the average heat capacity of part a with that of part b.
- In part c, for comparing C_p with those from nist database, create a column and compute the percent error of C_p, assuming that C_p from nist database is exact.

Exercise 4.5: For the function $x^2 \cos x$; $0 \le x \le 1$,

a. Calculate the numerical derivative of this function at each point using Δx values of 0.1 and 0.01. Be sure to use end-point formulas at $x = 0$ and $x = 1$. Compare the numerical derivatives at each point with the exact values.

b. Find the integral of this function using the trapezoidal rule over the same range. Calculate the % error of the integral at $x = 1$ for both Δx values.

Exercise 4.4: Evaluate the following integral using

a. The trapezoidal rule. Experiment with the Δx increment to produce good results.
b. Gauss quadrature. Use 2-, 3-, and 4-point formulas and compare results.

$$I = \int_0^1 \sqrt{x}\, dx$$

Exercise 4.1: The heat capacity at constant pressure is defined as

$$C_p = \left(\frac{\partial H}{\partial T}\right)_p$$

where C_p is the heat capacity at constant pressure, H is the molar enthalpy, and T is temperature. The following table shows heat capacity versus temperature data for carbon dioxide (http://webbook.nist.gov/chemistry/fluid/).

Temperature (°C)	Enthalpy (kJ/mol)
100	25.186
150	27.254
200	29.408
250	31.640
300	33.942
350	36.307
400	38.732
450	41.209
500	43.736
550	46.307

600	48.919
650	51.568
700	54.252
750	58.966
800	59.710

a. Use finite differences to compute the heat capacity of carbon dioxide at each of the given temperatures. Be sure to use the "end-point" formulas for the first and last entries.
b. Graph the enthalpy data and curve-fit it with a suitable polynomial. Then, calculate the heat capacity at each temperature using analytical differentiation. Compare these results with those of part a.
c. Compare the results for C_p with those from the *nist* database (these values appear in the data table of Exercise 4.2). Be sure to use consistent units for comparison.